

Improvement and Application of Grey Prediction Model for Road Traffic Accident

DAI Lei-lei

Traffic Management Research Institute of the
Ministry of Public Security
Wuxi, China

YU Chun-jun

Traffic Management Research Institute of the
Ministry of Public Security
Wuxi, China

GU Jin-gang

Traffic Management Research Institute of the
Ministry of Public Security
Wuxi, China

Abstract—In view of the disadvantages of GM(1,1) model, which has low precision for prediction of violent non-stationary data series. Based on availability and characteristics of accident data, the GM(1,1) improved model and Verhulst improved model are put forward by using metabolism methods. Application of traffic accident data from 1996~2009 of Heilongjiang Province shows that the Verhulst improved model is more precise than GM(1,1) improved model for violent changing data series. The average precision of Verhulst improved model can reach 85%, which can basically meet the demand for road traffic safety management.

Keywords—road traffic accident; grey prediction model; Verhulst model; metabolism method

I. INTRODUCTION

Road traffic accident is a random event, with its own contingency and ambiguity. The impact of road traffic accidents including human factors (drivers or traffic participants), vehicles, roads, traffic management and environment, some factors can be quantitatively described, but some factors are not precisely described [1]. If we looked the road traffic as a system, there are some identified factors such as road conditions, lighting conditions in this system, and there are also some uncertain factors, such as traffic accidents, traffic flow, weather conditions, drivers mental state [2]. Therefore, we can think a road traffic system as a grey system, and the grey system theory can be used for studying. Grey system theory is based on existing information on the system to predict the future. The essence of it is using historical data for the first-order accumulated generating operation, get a regular series after a strong build, and then to approximate exponential curve fitting of the generated series. Only small sample size of data needed of grey forecasting method to predict over other methods, and the precision has high accuracy, in particular the lack of basic data for the forecast better reflect its advantages.

II. ROAD TRAFFIC ACCIDENT GREY MODEL IMPROVEMENT

More common model of grey prediction is the GM (1,1) model [3,4], the model has a strong exponential law applicable to the data series, can only describe the monotonous process of change. It fitting poor and has low prediction accuracy to the

volatile data series[5]. But for the swing development of non-monotonic series or series with a saturation of the S-shaped, Verhulst model is more applicable. Verhulst model is mainly used to describe the process with saturation, which is the S-shaped course [6].

The forecast precision of these models is better while the data is not volatility. However, the traffic accidents have strong fluctuations and difficult to eliminate the impact of strong volatility data, so directly affect the prediction accuracy, and only at or near the stationary data series, which demonstrate monotonic or S-shaped swing development. Therefore, based on the availability of data, the metabolism methods were used for updating accident data series, and the GM (1,1) model and the Verhulst model were improved to reduce the traffic data out of volatile.

A. GM(1,1) Improved Prediction Model

The basic GM(1,1) model shown as:

$$\frac{dx^{(1)}}{dt} + ax^{(1)} = \mu \quad (1)$$

Where, a, μ are the parameters to be identified and internal variables in modeling process; $x^{(1)}$ is the new data series after accumulated process.

Parameter identification process is as follows:

① Constructed data matrix B

$$b = \begin{Bmatrix} -\frac{1}{2}[x^{(1)}(1) + x^{(1)}(2)] & 1 \\ -\frac{1}{2}[x^{(1)}(2) + x^{(1)}(3)] & 1 \\ \vdots & \vdots \\ -\frac{1}{2}[x^{(1)}(m-1) + x^{(1)}(m)] & 1 \end{Bmatrix} \quad (2)$$

② Constructed matrix vector y_m

$$y_m = [x^{(0)}(2), x^{(0)}(3), \dots, x^{(0)}(m)]^T \quad (3)$$

③ For the least squares calculator, obtained parameters a , μ

$$\hat{c} = \begin{bmatrix} a \\ \mu \end{bmatrix} = (B^T B)^{-1} B^T y_m \quad (4)$$

④ Constructed the time response function

$$\hat{x}^{(1)}(k) = [x^{(1)}(0) - \frac{\mu}{a}]e^{-a(k-1)} + \frac{\mu}{a} \quad (5)$$

⑤ Obtained the prediction value

Less accumulated value of the model (differential) operator, get the simulation prediction value of the original series:

$$\hat{x}^{(0)}(k) = \hat{x}^{(1)}(k) - \hat{x}^{(1)}(k-1) \quad (6)$$

⑥ Testing of the accuracy

The accuracy of prediction can test by the size of residuals and relative errors

The residuals can be calculated:

$$e(k) = x^{(1)}(k) - \hat{x}^{(1)}(k) \quad (k=1, 2, \dots, m) \quad (7)$$

The relative errors can be calculated:

$$e'(k) = \frac{|e(k)|}{x^{(1)}(k)} \times 100\% \quad (8)$$

⑦ Model improved

Grey prediction model with high precision is only a few data around the origin. In general, the more away from the time of the origin, the more there is no predictive value. In practice, we must continue to consider the disturbances entered into those systems over time, to keep each new data into the model anytime [7].

To ensure the data for real-time, we can use metabolic methods to improve the GM (1,1) model accuracy by updating history data. The method is removing one of the oldest messages and insert into one of the latest information [8], the original data modeling is listed as:

$$x^{(0)} = [x^{(0)}(2), \dots, x^{(0)}(m), x^{(0)}(m+1)] \quad (9)$$

By taking the data into the above model, calculation steps and formulas remain unchanged.

B. Verhulst Improved Prediction Model

When Verhulst model is used for predicting, it first through the AGO (1-AGO) treatment, and weaken the randomness of the new data series [9], the establishment of differential equations is established as:

$$\frac{dx^{(1)}}{dt} + a_{\text{verhulst}} x^{(1)} = u_{\text{verhulst}} (x^{(1)})^2 \quad (10)$$

Model construction process is as follows

① Constructed data matrix B_{Verhulst}

$$z^{(1)} = \begin{bmatrix} -(x^{(1)}(1) + x^{(1)}(2))/2 \\ -(x^{(1)}(2) + x^{(1)}(3))/2 \\ \dots \\ -(x^{(1)}(m-1) + x^{(1)}(m))/2 \end{bmatrix} = \begin{bmatrix} z^{(1)}(2) \\ z^{(1)}(3) \\ \dots \\ z^{(1)}(m) \end{bmatrix}$$

$$B_{\text{Verhulst}} = \begin{bmatrix} -z^{(1)}(2) & (z^{(1)}(2))^2 \\ -z^{(1)}(3) & (z^{(1)}(3))^2 \\ \dots & \dots \\ -z^{(1)}(m) & (z^{(1)}(m))^2 \end{bmatrix} \quad (11)$$

② Constructed matrix vector

$$Y_{\text{Verhulst}} = [x^{(1)}(2) - x^{(1)}(1), x^{(1)}(3) - x^{(1)}(2), \dots, x^{(1)}(m) - x^{(1)}(m-1)]^T \quad (12)$$

③ Obtained parameters vectors by using the least squares calculator

$$\begin{aligned} \lambda_{\text{Verhulst}} &= [a_{\text{Verhulst}}, u_{\text{Verhulst}}]^T \\ &= (B_{\text{Verhulst}}^T B_{\text{Verhulst}})^{-1} B_{\text{Verhulst}}^T Y_{\text{Verhulst}} \end{aligned} \quad (13)$$

④ Constructed the time response function, and restore the predictive value

The time response function is:

$$\begin{aligned} \hat{x}_v(k+1) &= a_{\text{Verhulst}} x^{(1)}(0) / [u_{\text{Verhulst}} x^{(1)}(0) \\ &\quad + (a_{\text{Verhulst}} - u_{\text{Verhulst}} x^{(1)}(0)) e^{a_{\text{Verhulst}} k}] \end{aligned} \quad (14)$$

The predictive value is:

$$\hat{x}^{(0)}(k) = \hat{x}_v^{(1)}(k) - \hat{x}_v^{(1)}(k-1) \quad (15)$$

⑤ Testing of the accuracy

The residuals can be calculated:

$$e_v(k) = x^{(1)}(k) - \hat{x}_v^{(1)}(k) \quad (k=1, 2, \dots, m) \quad (16)$$

The relative errors can be calculated:

$$e'(k) = \frac{|e_v(k)|}{x^{(1)}(k)} \times 100\% \quad (17)$$

⑥ Model improved

The original data modeling is constructed by using metabolic methods, the method is removing one of the oldest messages and insert into one of the latest information, it is listed as:

$$x^{(0)} = [x^{(0)}(2), \dots, x^{(0)}(m), x^{(0)}(m+1)] \quad (18)$$

By taking the data into the above model, calculation steps and formulas remain unchanged.

III. CASE STUDIES

According to the traffic accidents statistical data from 1996 to 2009 of Heilongjiang Province, it's from the Annual Traffic Statistics Report of Heilongjiang Province. The prediction

accuracy of traffic accidents is tested for the improved models based on grey prediction, the original data shown in Table 1.

TABLE I. THE TRAFFIC ACCIDENTS STATISTICAL DATA FROM 1996 TO 2009 OF HEILONGJIANG PROVINCE

Year	Traffic accidents (cases)	Year	Traffic accidents (cases)
1996	2589	2003	9245
1997	2691	2004	9163
1998	2823	2005	7552
1999	3092	2006	6732
2000	12091	2007	6716
2001	12378	2008	4528
2002	11471	2009	3331

TABLE II. THE ERRORS COMPARISON OF IMPROVED GREY MODELS FOR TRAFFIC ACCIDENTS PREDICTION

Year	Traffic accidents predicted (cases)				Year	Traffic accidents predicted (cases)			
	GM(1,1) improved model	Relative errors	Verhulst improved model	Relative errors		GM(1,1) improved model	Relative errors	Verhulst improved model	Relative errors
1996	2589	0.00%	2589	0.00%	2003	15144	63.81%	10151	9.80%
1997	8372	211.11%	2242	16.69%	2004	16254	77.39%	7141	22.07%
1998	3627	28.48%	3953	40.03%	2005	19321	155.84%	6438	14.75%
1999	4827	56.11%	3592	16.17%	2006	21508	219.49%	5107	24.14%
2000	6424	46.87%	9523	21.24%	2007	23315	247.16%	6273	6.60%
2001	8550	30.93%	11878	4.04%	2008	26021	474.67%	3270	27.78%
2002	11379	0.80%	12150	5.92%	2009	28456	754.28%	3762	12.94%

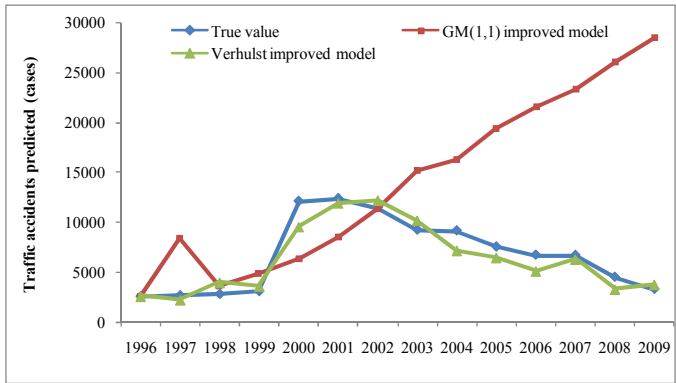


Figure 1. The results analysis of grey improved models for traffic accidents prediction

From the results comparison of Table 2 and Figure 4, the GM(1, 1) improved model has larger error for traffic accident prediction, the average prediction error can reach to 169.07%. In contrast, the Verhulst improved model has high precision, and the average prediction error is only 15.87%. Taking into account the randomness of traffic accidents are relatively strong, the prediction accuracy about 85% is acceptable. Therefore, the Verhulst improved model has better fitting results for traffic data with strong volatility.

The traffic accidents from 1996 to 2004 of Heilongjiang Province are used for model calibration data, and traffic accidents from 2005 to 2009 are predicted. Through model simulation, GM (1,1) model parameters $a = -0.286$, $\mu = 8728.717$. Verhulst model parameters $a_v = -0.655$, $\mu_v = -0.000009$, the time response function are:

$$\hat{x}^{(1)}(k) = 33125.041e^{0.286(k-1)} - 30536.041 \quad (19)$$

$$\hat{x}_v^{(1)}(k+1) = 1/(0.000013 + 0.00037e^{-0.655k}) \quad (20)$$

The data series is updated by using metabolism methods, the prediction results of GM(1,1) improved model and Verhulst improved model are shown in Table 2, the comparisons of prediction values and true values are shown in Figure 1.

IV. CONCLUSIONS

Because the common grey models have shortcomings for road traffic accident prediction, two improved Grey prediction models are proposed: GM (1,1) improved model and Verhulst improved model. With the analysis results of case studies, it showed the Verhulst improved model with better adaptation for traffic accident data with strong randomness and more intense changes.

As road traffic accidents are random, volatility and uncertainty, although improved grey models for road traffic accident have increased prediction accuracy, but we should further in-depth research various prediction methods for the purpose of effective traffic accident prevention and control, so as to improve the accuracy of road traffic accident prediction.

REFERENCES

- [1] Wu X B. "Application of residual error grey forecast model in traffic accident forecast," Technology & Economy in Areas of Communications, 2001, (1), pp.33-34.
- [2] Wang L L, Shi Z K, Hu T P. "A real-time predict model for freeway traffic accidents basing on fuzzy judgment," Systems Engineering—Theory & Practice, 2003, 23(5), pp.141-144.
- [3] Liu J Q, Chen L, Liu J W. "The grey forecasting GM(1,1) model in forecasting traffic accident," Guangxi Communications Science and Technology, 2003, 28(4), pp.106-109.

- [4] Ji S F. "Optimal replenishment policies for deteriorating items based on the forecasting method of grey model," *Systems Engineering—Theory & Practice*, 2008, 28(7), pp.93-99.
- [5] Ma Y L, Pei Y L. "Grey model based on multi-factors for forecasting road accidents," *Central South Highway Engineering*, 2007, 32(3), pp.147-150.
- [6] Liu S F. "Grey system theory and its application," Beijing: China Science Press, 1999, pp.120-125.
- [7] Yu H J, Zhang X D. "The application of dynamic equal dimension new information model in prediction of the traffic accidents," *Yiqiyibiao Yonghu*, 2008, 115(2), pp.49-50.
- [8] Hu K. "Study on method and application of grey prediction evaluation," *Nanjing University of Aeronautics and Astronautics*, 2004, pp.14-16.
- [9] Cao C W, Gu X S. "The application of grey dynamic Verhulst metabolism model in the prediction of products' prices and demands," *Information and Control*, 2005, 34(4), pp.398-402.